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Eating breakfast enhances the efficiency of neural networks engaged during mental arithmetic in school-aged children

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ABSTRACT

To determine the influence of a morning meal on complex mental functions in children (8–11 y), time–frequency analyses were applied to electroencephalographic (EEG) activity recorded while children solved simple addition problems after an overnight fast and again after having either eaten or skipped breakfast. Power of low frequency EEG activity [2 Hertz (Hz) bands in the 2–12 Hz range] was determined from recordings over frontal and parietal brain regions associated with mathematical thinking during mental calculation of correctly answered problems. Analyses were adjusted for background variables known to influence or reflect the development of mathematical skills, i.e., age and measures of math competence and math fluency. Relative to fed children, those who continued to fast showed greater power increases in upper theta (6–8 Hz) and both alpha bands (8–10 Hz; 10–12 Hz) across sites. Increased theta suggests greater demands on working memory. Increased alpha may facilitate task-essential activity by suppressing non-task-essential activity. Fasting children also had greater delta (2–4 Hz) and greater lower-theta (4–6 Hz) power in left frontal recordings—indicating a region-specific emphasis on both working memory for mental calculation (theta) and activation of processes that suppress interfering activity (delta). Fed children also showed a significant increase in correct responses while children who continued to fast did not. Taken together the findings suggest that neural network activity involved in processing numerical information is functionally enhanced and performance is improved in children who have eaten breakfast, whereas greater mental effort is required for this mathematical thinking in children who skip breakfast.

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1. Introduction

The importance of mathematical competence for effective functioning in today's society has led to intensified research efforts to identify the behavioral and neurocognitive processes and factors that influence the development and expression of mathematical skills. This research has revealed that the foundations for numerical representation are evident in infants [1,2] and develop further through exposure to mathematical information embedded in the everyday environment prior to the introduction of structured mathematical education [3–5]. Critical factors in this development are basic mathematics-related skills, such as number sense, size or quantity discrimination, and ability to add and subtract simple whole numbers, as well as cognitive abilities involving memory functions and processing speed (reviewed in [6]). There is a general consensus that in children and adults a network of frontal and parietal brain regions are activated during calculation tasks and that with increasing mathematical expertise there is a shift in emphasis from frontal to parietal processes [7–9].

This development is also influenced by environmental variables and individual capability as reflected by SES and IQ [6,10]. However, a fundamental factor not generally considered in studies involving healthy participants is their nutritional status at the time of assessment. This is surprising in view of the undisputed fact that brain function is dependent on adequate nutrition, and that short-term variations in the amount and composition of nutrient intake in healthy individuals influence measures of cognitive function [see [11–13] for reviews]. For example, cognitive functions, including those involved in mental arithmetic [12], have been shown to be sensitive to variations in nutrient intake, levels of glucose in the blood, interactions of glucose with other nutrients, e.g., proteins and fats [14,15] and hormones involved in glucose regulation, e.g., cortisol [16].

Cognitive functioning is likely to be more susceptible to short-term variations of nutrition in children because of their high metabolic rates and increased energy demands [17,18]. On a daily basis the greatest nutritional variation occurs during overnight fasting, which may last more than 9–10 h in children because of their extended sleep time [19]. Nutrients provided by breakfast are therefore particularly important for meeting the physiological and cognitive demands children face in the morning. Immediate cognitive benefits of eating breakfast that have been reported in children include improved memory, attention, and

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visual discrimination [20–23]. Studies regulating breakfast eating over longer time periods (days or weeks) have reported improved academic performance on complex mental tasks such as arithmetic [24,25]. However, the neural correlates of these nutrition-related effects remain largely undefined, and there are no studies comparing measures of brain activity recorded during the performance of mental arithmetic in school-aged children who had eaten or skipped breakfast.

In the present study we used time–frequency analysis to document covariations among low frequency EEG activities occurring while children who were fasting or had eaten breakfast performed time-regulated mental computations of simple addition problems. Recordings from placements over frontal and parietal brain areas involved in mathematical cognition [26–28] were analyzed with a focus on activity from 2 to 12 Hz. These frequencies were selected because of the demonstrated relevance of EEG activities in this range to attentional and memory processes involved in mathematical thinking [29–31]. The 2–12 Hz frequency range includes traditionally defined delta (2–4 Hz), theta (4–8 Hz) and alpha (8–12 Hz) bands of activity. Although the cognitive and behavioral correlates of these activities are often based on these broadband definitions, it is clear that sub-bands within each frequency grouping may have different functional correlates. In this regard, lower and upper frequencies in the theta and alpha bands have received the greatest attention. Theta activities have been most consistently associated with memory-related functions, with increased lower frequency activity (4–6 Hz) associated with attention, encoding, retrieval, memory load, and semantic memory processes [32–35], and increased upper frequency activity (6–8 Hz) reflecting site-specific memory maintenance and memory load influences [36,37].

Differential effects associated with low (8–10 Hz) and high (10–12 Hz) alpha frequencies have been more extensively documented. Low alpha effects are topographically diffuse and considered to reflect general, non-specific variations in attention and arousal and high alpha effects more topographically restricted and task-specific and semantic memory processes [38]. Lower and upper alpha bands are also differentially sensitive to memory load and task difficulty [37,39]. These sub-band correlates are generally associated with condition-related alpha suppression (desynchronization) which is considered to reflect excitatory processes [40].

Recently it has been proposed that stimulus-related increases (synchronization) in alpha activity enable the processing of task-relevant information by inhibiting task-irrelevant neuronal activity [40,41]. Importantly, these formulations of theta and alpha functions are considered to represent processing modes flexibly tailored to meet task-specific requirements and demands, rather than mechanisms that are domain distinct. Given these considerations it is not surprising to find overlaps in the functional correlates of these bands and sub-bands since these correlates result from interactions among frequency bands, task-relevant sites, and task-related determinants [42–48].

Slower memory recall in children who skip breakfast [22,49] indicates that working memory is sensitive to fasting, and working memory is considered pivotal to mathematical learning and performance in children [50,51]. In view of the association of EEG theta activity with working memory and findings of increases in this activity with increased mental effort or memory load, we expect *fasting* children will show greater theta power than *fed* children. Furthermore, since the ability to inhibit task-irrelevant information from gaining access to working memory is critical to working memory function and efficient mathematical performance [50] and such inhibitory influences have been associated with increases in delta [29] and alpha activity, we hypothesize that power in these bands will be greater in *fasting* than *fed* children. It is expected that in children who eat breakfast these relatively reduced levels of 2–12 Hz brain activity will be associated with improved task performance. The requirement of lower brain activity to successfully meet task demands is consistent with the concept of neural efficiency as proposed by Haier et al. [52] to characterize the reduced brain energy consumption (glucose metabolism) required by good relative to poor performers on a complex task. Evidence in support of this concept has been provided by studies measuring brain electrical activity and metabolism (see [53–55]). The overarching hypothesis, then, is that morning nutrition will increase the efficiency of brain processes involved in a complex mental function.

2. Methods

2.1. Participants

Study participants were recruited from the local community. Participants were 8–11 years old, above the 25th percentile in weight-for-age and height-for-age, between the 25th and 95th percentiles in BMI, and scored at least 80 on IQ evaluations (Wechsler Abbreviated Scale of Intelligence; Table 1). Children having medical conditions or taking medications that would prevent the administration of the study protocol or that were known to influence study variables were excluded from participation. Of 116 children who met these criteria and completed the study protocol, eighty-one healthy, right-handed children (44 females) are included for this report. The remaining 35 were excluded for the following reasons: 16 were left handed (only right-handed children were included in this report to avoid confounds in hemispheric analyses; handedness was determined from parental and children's self-reports and the preferred hand was used for motor responses on study tasks); 17 showed excessive artifact in their recordings resulting from movement or blinking; and 2 failed to follow testing instructions.

Three day food records and information about children's breakfast eating habits (e.g., frequency and composition) were provided by parents. Assessments of general math skills and performance were made using the Wide Range Achievement Test-3 (Arithmetic subtest standard scores) and a Math Fluency Task (20 simple addition and subtraction problems to be completed in ≤10 min [56]) (see Table 1). The study protocol was approved by the University IRB Committee. Informed consent was obtained from parents and assent from the children.

2.2. Procedures

Participants and one of their parents stayed at the research facility the night prior to morning assessments, reporting to the facility at

Table 1
Comparisons between treatment groups on background study measures. Table values are means ± SD. Significant differences are bolded.

	Group		
	Fasting		Fed
	(n = 41; 19 male)	p	(n = 40; 18 male)
Age (yr)	9.78 ± 0.83	.97	9.78 ± 0.82
Weight (kg)	36.2 ± 5.4	.56	37.0 ± 7.0
Height (cm)	140.5 ± 6.2	.49	141.5 ± 7.3
BMI	18.3 ± 2.0	.92	18.4 ± 2.2
BMI percentile	67.2 ± 20.5	.87	66.5 ± 21.0
IQ	113.0 ± 9.8	.54	111.3 ± 15.3
SES	46.9 ± 11.5	.60	44.7 ± 14.2
WRAT Arithmetic	108.2 ± 10.3	**.011**	101.6 ± 12.7
Math fluency — Score	18.9 ± 2.2	.12	18.1 ± 2.3
Math fluency — Time (s)	133.2 ± 84.6	.30	151.9 ± 77.8
Total sleep (min)	492.3 ± 44.3	.19	506 ± 42.8
Sleep efficiency	90.9 ± 6.6	**.005**	94.8 ± 4.6
Blood glucose 1	88.0 ± 8.9	.31	86.2 ± 7.4
Blood glucose 2	87.4 ± 8.3	**<.001**	113.3 ± 16.4
Blood glucose 3	82.0 ± 6.5	**<.001**	92.6 ± 11.3

BMI (body mass index); IQ (intelligence quotient — based on Wechsler Abbreviated Scale of Intelligence); SES (social economic status — based on [92] Four-Factor Index of Social Positions); WRAT (Wide Range Achievement Test-3).

approximately 5:00 p.m. for orientation to the laboratory and assessment procedures. Orientation included the administration of a shortened version of the math paradigm to familiarize participants with the problem presentation rate and response demands of the paradigm. Problems used in this practice session were similar to those in the experimental paradigm but did not include the specific problems in that paradigm. After these procedures children ate dinner and relaxed until going to bed no later than 9:00 p.m. Only water was allowed after bedtime. A nurse monitored the children during the overnight stay and ensured that fasting and bedtime requirements were observed. Children wore a wrist actigraph overnight (Mini-Motionlogger, Ambulatory Monitoring, Inc.) to provide information regarding the amount and quality of sleep. Since those assigned to the fasting condition would not be eating until lunch time on the testing day, blood glucose levels were assessed via finger stick (Glucometer Elite XL®, Bayer Corporation) as a safety precaution upon awakening in the morning and again following the mid-morning break and just prior to lunch.

Shortly after participants were awakened at 6:45 a.m., electrodes were attached for recordings of EEG [tin electrodes in an elastic cap (Electro-Cap International, Inc.); F3, F4, C3, C4, P3, P4, O1, O2 and right mastoid all referenced to left mastoid placement] and eye movements (bipolar EOG from placements at the outer canthi and above and below one eye for recording horizontal and vertical eye movements, respectively). The ground electrode was located on the shoulder. Impedances were maintained at <10 KΩ. Physiological signals were filtered (0.1 Hz high pass, 100 Hz low pass), amplified (EEG: gain 20,000; EOG: gain 1000), and stored for offline analysis.

Beginning at 8:00 a.m. while still fasting and seated comfortably in a dimly lighted room, the children began performing a battery of tasks which included the mental arithmetic paradigm. With scheduled rest periods, this initial testing session lasted until 9:30 a.m. At the end of this session a 40 min break occurred during which children were taken to another area of the research center and randomly assigned to *fed* or *fasting* treatment groups. Participants then either ate breakfast ($n = 40$, 18 males) or continued fasting ($n = 41$, 19 males) and rested until the end of the break period.

The standardized breakfast, based on USDA School Breakfast Program minimum requirements, consisted of dry cereal (3/4 cup unsweetened), milk (2% fat, 8 fluid ounces), toast (1 slice made from enriched white bread, plain), and applesauce (1/2 cup, unsweetened). This breakfast contained 14 g protein, 6 g fat, and 57 g carbohydrate (340 cal: 17, 16, and 67% from protein, fat and carbohydrate, respectively) [57]. For sedentary preadolescents in the 8–11 years age range, the daily estimated energy requirement (EER; averaged across age and gender) is ~1485 kcal [58]. This breakfast serving therefore provided ~23% of the daily EER for these subjects, and age-specific values were similar for 8, 9, 10, and 11 year-old children, i.e., 24%, 23%, 23% and 21%, respectively.

Following the break, the children returned to the laboratory and repeated the test battery. Order of tasks within each session was randomized across subjects, but within subjects the same order was maintained across sessions. Those conducting the recordings were unaware of the children's treatment conditions. In this report the two testing sessions will be referred to as Phase 1 (all children fasting) and Phase 2 (either continuing to fast or having eaten breakfast).

The mental arithmetic task was time-regulated and involved computations of 150 simple problems (75 addition, 75 subtraction, randomized). The current report is focused on outcome measures (number correct and response time) and EEG correlates associated with correctly answered addition problems only. These problems used single and double digit numbers and none of the computations required a carry operation. Each problem was presented for 200 ms on a monitor positioned 1.5 m from the child followed 1 s later by the simultaneous display of three answers (200 ms; labeled a, b, and c), one of which was correct. Children sat with their index, middle and ring fingers of their right hand each resting on a response button. They were required to press the response button corresponding to their choice of the correct answer within 1.75 s of answer presentation offset and earned 2 cents for each correct response. Feedback was provided at the end of the response window (a smiling face if correct and a frowning face if incorrect). The interval from the onset of one problem to the onset of the next was 5.2 s. Including one scheduled 2 min break, the paradigm required 15 min to complete.

2.3. Data processing and quantification

EEG recordings were digitized at 1024 Hz. All channels were re-referenced offline to the mean value of the mastoids and then screened for blinks (250 μV threshold applied to the fast running average of the vertical EOG recording) and other high amplitude artifact (≥30 μV on EEG channels). The data were then segmented into 1200 ms trials beginning 200 ms before, and ending 1000 ms after, stimulus onset. Data were then down-sampled to 200 samples per/s prior to further analysis. A continuous wavelet transform (Morlet) [Matlab (Version R2009b) Signal Processing Toolbox (The Mathworks, Inc.)] was then applied to single artifact-free trials from bilateral frontal (F3, F4) and parietal (P3, P4) sites. The wavelet transformation was performed on the 1200 ms EEG segments in 0.25 Hz steps. The results, expressed as dB relative to 1 μV^2, were then averaged into 2 Hz bands. Data from the initial and final 200 ms were eliminated to reduce edge effects leaving wavelet data for the 800 ms following problem presentation onset.

Wavelet analysis was used since it permits examination of changes in spectral content of EEG signals over time, and therefore might reveal group differences in mental arithmetic that unfolded across time. Inspection of the curves of activity for each frequency band across the 800 ms showed that group differentiation was primarily reflected in sustained level differences across time. Therefore, to simplify data analysis and presentation, statistical analyses were based on wavelet data averaged across this 800 ms interval for each of the frequency bands of interest, i.e., delta (2–4 Hz); lower (4–6 Hz) and upper (6–8 Hz) theta; and lower (8–10 Hz) and upper (10–12 Hz) alpha bands.

2.4. Statistical analyses

Analysis of variance (ANOVA) and t-tests were performed to compare groups on each background variable. Time–frequency data were analyzed using multivariate and repeated measures ANOVAs and post-hoc t-tests. Greenhouse–Geisser corrections were made for violations of sphericity. Covariates were also included to evaluate the influence of variables relevant to mathematical competence (i.e., age, WRAT-3 Arithmetic Scores, Math fluency scores) on brain activity measures. Data were analyzed using SPSS Statistics 17.0. A value of $p \leq .05$ was considered statistically significant. Comparisons where $.05 < p < .10$ are reported to indicate possible trends.

3. Results

3.1. Background measures

Study groups were well-matched on inclusion criteria, i.e., there were no significant between-group differences on age, IQ, or BMI percentile (Table 1). However, children assigned to the *fasting* group did have higher WRAT-3 Arithmetic subtest scores than those in the *fed* group ($p = .01$).

As determined from food records and information regarding daily eating habits, all but one child in each group typically ate breakfast. In the laboratory 90% (36/40) of the children in the *fed* group consumed all of the standardized meal, and the remaining 4 children consumed 68%, 81%, 92%, and 97% of the meal. Protocol procedures and testing schedules were maintained as planned throughout the investigation.

Overnight actigraph recordings were not obtained for eight children (four in each group) because of technical problems. Group comparisons for the remaining children showed similar amounts of total sleep time, with both groups averaging more than 8 h of sleep. Sleep efficiency (time asleep/total bed time) was greater in *fed* children (fasting: 90.9%; fed: 94.8%; $p=.005$), but since sleep data were not obtained for all children this measure was not included as a covariate. However, because of the potential impact of this behavior on daytime performance—especially in children where sleep requirements are greater—analyses were conducted which included sleep efficiency among covariates by inserting group averages for missing sleep data. The results did not alter group-related effects observed when this measure was not considered, and no group effects related to sleep efficiency emerged.

Blood glucose levels upon awakening were in the normal range and similar across groups. Blood glucose increased significantly within 30 min of eating in *fed* children relative to their initial fasting levels ($p<.001$) and compared to those who continued to fast ($p<.001$). Levels remained higher in *fed* than *fasting* children when measured at lunch time about 2 h after eating breakfast ($p<.001$; Table 1).

3.2. Performance: correct responses and response time

Within testing phases groups did not differ significantly in the number of or latency to (response time) correct responses. Across test phases, children in the *fed* group showed an improvement in the number of correct responses ($p=.001$), and children in the *fasting* group showed a decrease in response time ($p=.01$; Fig. 1).

3.3. Time–frequency analyses

Group averages of frequency specific power responses during the performance of mental addition are presented as a function of scalp region and test phase in Fig. 2. Effects common to both groups across test periods included: greater power in the right than the left hemisphere; greater power in frontal than parietal regions for 2–8 Hz activity; and greater power in parietal than frontal regions for 8–12 Hz activity. With the exceptions of hemispheric comparisons for delta and broad band theta activity (4–8 Hz) and frontal–parietal comparisons for upper theta (6–8 Hz), these power differences were statistically significant [F (1, 79) values ranging from 9.88 to 258.72, all $p<.01$]. Across bands and sites, power increased significantly from Phase1 to Phase 2 [F (1, 79) values of 7.66 to 97.68, all $p<.01$]. Significant interactions involving group and phase variables included complex four-way interactions (hemisphere by frontal–parietal by phase by group) for delta [F (1, 79) = 5.71, $p=.019$] and lower theta [F (1,79) = 5.07, $p=.027$], as well as phase by group interactions for upper theta [F (1,79) = 10.64, $p=.002$] and lower alpha [F (1,79) = 9.96, $p=.002$]. When covariates were considered, statistical significance was maintained for findings of greater frontal than parietal delta power across groups and phases [F (1,76) = 6.75, $p=.011$], and for all the treatment-related interactions noted above, i.e., four-way interactions for delta [F (1,76) = 10.55, $p=.002$] and lower theta [F (1, 76) = 6.20, $p=.02$] and phase by group interactions for upper theta [F (1,76) = 9.85, $p=.002$] and lower alpha [F (1, 76) = 8.10, $p=.006$].

To focus on within-group treatment effects, analyses with adjustments for covariates were performed on the difference between Phase 2 and Phase 1 power values. These data are presented in Fig. 3. In the lower frequency bands significant group-related effects were found for hemisphere by frontal–parietal by group interactions [delta: F (1, 76) = 11.84, $p=.001$; lower theta: F (1, 76) = 5.57, $p=.02$]. For delta activity this interaction reflected lower left frontal power increases in *fed* children compared to left parietal ($p=.03$) and right frontal($p<.05$) increases for this group, and to left frontal values for the *fasting* group ($p=.07$). Examination of the interaction for lower theta also showed a left frontal emphasis, but here power increases for this band in *fasting* children were higher than that in either the left

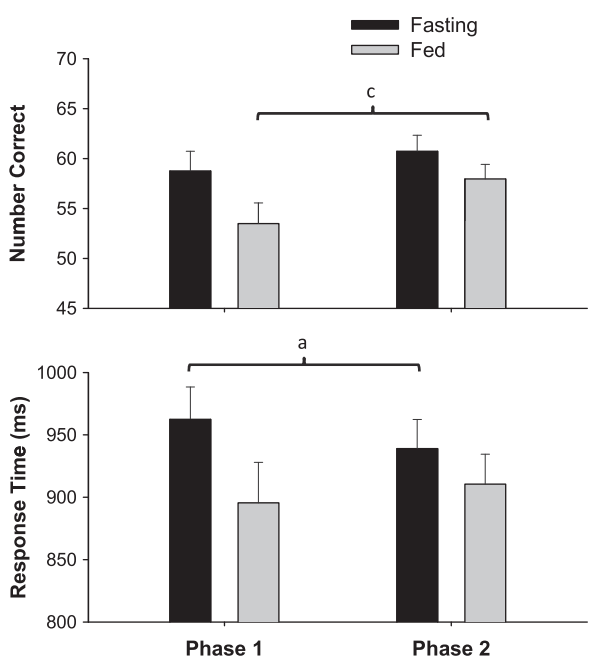

Fig. 1. Group comparisons of math paradigm performance measures. Upper panel shows number correct out of 75 possible. Lower panel shows response time to selection of answer. Graphed values are means + SE. Statistically significant differences are indicated (a: $p<.05$; c: $p<.001$).

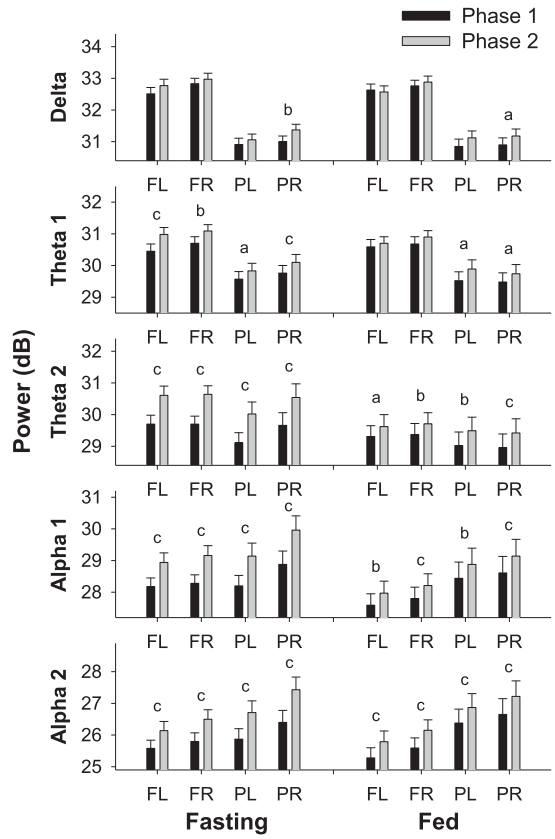

Fig. 2. Group power values during mental calculation. Individual frequency band data (800 ms averages + SE) are graphed for left hemisphere frontal (FL) and parietal (PL) sites and right hemisphere frontal (FR) and parietal (PR) sites for each test phase. Significant between-phase differences are indicated (a: $p<.05$; b: $p<.01$; c: $p<.001$).

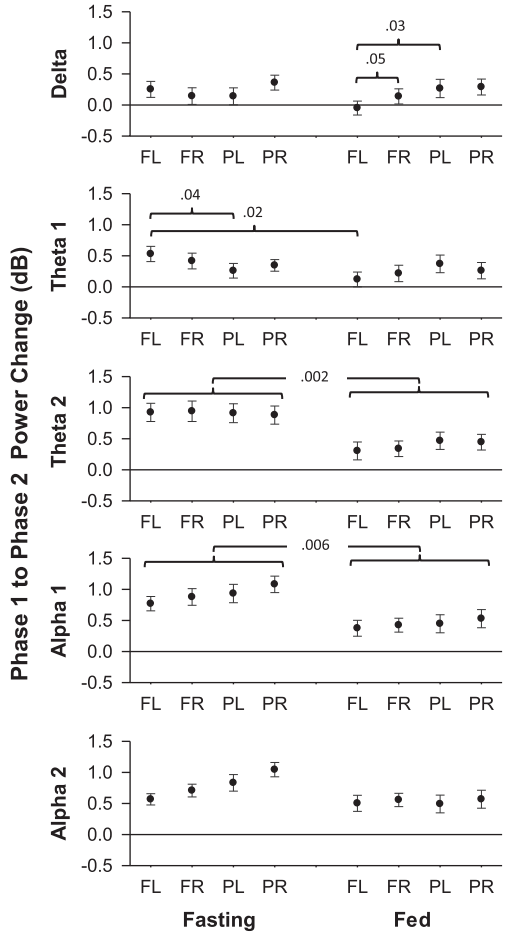

Fig. 3. Treatment effects based on between-phase changes in power. Group means of Phase 2 relative to Phase 1 values (800 ms averages + SE) are presented for individual frequency bands for left hemisphere frontal (FL) and parietal (PL) sites and right hemisphere frontal (FR) and parietal (PR) sites. Significant within- and between-group effects are indicated by brackets with associated probability values. Probability values for Theta 2 and Alpha 1 indicate group main effects.

parietal region in this group ($p = .04$) or left frontal region in *fed* children ($p = .02$).

In the higher frequency bands group main effects reflected greater power increases in *fasting* than *fed* children [upper theta: $F(1, 76) = 9.98$, $p = .002$; lower alpha: $F(1, 76) = 7.96$, $p = .006$; and with a trend for upper alpha: $F(1, 76) = 3.12$, $p = .08$].

4. Discussion

4.1. General

This study evaluated the effects of a morning meal on complex mental functions involved in solving arithmetic problems in 8–11 year old children. To make this determination EEG activity was recorded from placements over frontal and parietal regions important for mathematical functions while children performed mental calculations of simple addition problems, first after overnight fasting and again after they had either eaten or continued fasting. Frequency-specific EEG measures were adjusted for variables that influence or reflect the development of mathematical skills, i.e., age and measures of mathematical proficiency. As hypothesized, the findings indicate that the efficiency of neural networks engaged during mathematical processing was enhanced in children who had eaten breakfast compared with those who continued fasting. Furthermore, these effects were associated with group-related variations in performance measures, i.e., an increase in number of correct responses in *fed* children and faster response times in *fasting* children. These findings are discussed below.

4.2. Variations in regional EEG during mental arithmetic

This is the first study to apply wavelet analyses to brain activity generated during mental arithmetic in children, and the first to consider how variations in nutrition may modulate these correlates of mathematical thinking. The few studies of mental calculations in adults using this method of analysis have involved more complicated mathematical operations, i.e., two-digit multiplication [59], continuous mental arithmetic [60], or multiple operators involving addition and subtraction [61], and have not included measures of nutritional status. However, comparisons of brain activity during mental arithmetic in children and adults using various measures [e.g., EEG [31], event-related potentials [62], and imaging [7]] indicate that neural networks underlying mathematical thinking are similar in children and adults, and that the development of mathematical skills with experience is reflected by changes in problem-solving strategies associated with differences in levels of activation within these networks [7,63].

Results of the current study are in general agreement with these developmental observations. For example, given the reported inhibitory influences associated with delta and alpha activity [29,62], the observed greater right hemisphere power for these bands would promote the left hemisphere emphasis reported for processing of mathematical information [7,28,64] by reducing neuronal activity in areas that are not necessary for, or that might interfere with, task demands. The finding of greater power at frontal than parietal sites in lower theta is consistent with the recognized role of theta activity in frontal processes activated in tasks like mental arithmetic that are dependent on working memory [45,64,65], and with the greater reliance of children on frontal than parietal functions while mathematical skills are developing [8,9,28,51]. It is important to note that these nutrition-related influences are considered to be mediated by effects on processes fundamental to information processing and behavior, e.g., attention and working memory, which are tailored to meet task-demands and in this regard are not specific to mental arithmetic. For example, both common nutrition-related effects and task-specific differences are evident between the findings of the present study and those of Pivik and Dykman [21] who used ERS/ERD methodology to study the effects of morning nutrition on cued visual continuous task performance (CPT) in *fasting* and *fed* children. In both studies *fed* children showed less low alpha synchronization than *fasting* children, but interestingly faster response times were observed in *fed* children in the CPT study, but in *fasting* children in the current study. The different group response time effects across studies may reflect both task-specific and methodological factors.

4.3. Nutritional status, regional EEG and performance

Evaluating the effects of eating or skipping breakfast on mental arithmetic using frequency band and brain region related difference scores between Phase 1 and Phase 2 provided a normalized index of treatment influences on EEG measures. These results, adjusted further for age and measures of math ability, showed that power levels in Phase 2 were either maintained or increased relative to those in Phase 1 for both groups. These group-common effects may reflect a combination of factors, including circadian variations in 2–12 Hz activity across the morning related to increasing arousal and alertness [66] and increased attention and effort required by extended testing [67] and mental fatigue [68]. Similar influences were advanced to explain increases in alpha activity across the morning noted in *fasting* and *fed* children [21].

Group differences in these between-phase comparisons identify brain regions and indicate frequency-related neural networks involved in mathematical thinking that are sensitive to transient variations in nutritional status in children. The group differences – *fasting* relative to

fed children – included greater increases in power for upper theta and both alpha bands across sites and for delta and lower theta at the left frontal site. Power increases in the higher frequency bands (6–12 Hz) have been related to increased working memory [theta: [31,32,36]] and to the facilitation of task-essential activity by decreasing activity in regions not necessary for task performance [alpha bands: [44,69]].

The presence of group differences in the left frontal recordings is consistent both with the involvement of this region in attention, working memory, and executive function processes important for mental arithmetic [8,70] and with reports that increased activation of frontal processes in children performing mental calculations is particularly prominent while mathematical skills are still developing [28,71].

Studies in adults suggest that the greater left frontal delta and lower theta power in the *fasting* group would work to amplify the higher frequency (6–12 Hz) effects noted above. These studies indicate that the increased delta activity may reflect greater activation of processes that promote attention to internal processing during performance by suppressing interfering activity [29,30]. The increased theta activity may represent greater demands on working memory for this mental function [57,72,73].

These results are also relevant to strategy-related modulation of EEG activity during mental arithmetic. Although children in this study were not questioned regarding strategies used in problem solving, the consensus is that procedural methods (e.g., counting and transformation) are used during the development of mathematical skills and direct memory retrieval becomes more prevalent with greater development of these skills and working memory capabilities [74,75]. Studies based on ERS/ERD methods in adults showing increased theta activity in association with computation of simple addition problems have attributed this to memory retrieval processes [72,76]. Since children do not rely heavily on direct retrieval of rote knowledge for problem solving, the observed theta increases likely indicate greater mental effort associated with working memory processes as opposed to increases associated with direct memory retrieval.

Studies in adults [72,76] have also reported reduced alpha activity during computation of simple mathematical problems—unlike the increased alpha activity observed in the present study. While these age-related effects may in part reflect differences in methods of analysis (ERS/ERD vs. treatment-related differences), it may also be that with development the efficiency of neural systems involved in processing mathematical information decreases the extent to which alpha-related suppressive effects are required. It remains unclear, however, whether similar dietary variations in adults would provoke the frequency-related processing relationships observed in children in this study.

Considered together, the observed general and region-specific findings suggest that brain processes involved in mathematical thinking are facilitated in children who eat breakfast compared with those who skip breakfast. The fasting-related effects did indicate increases in functions that serve to enhance information processing, i.e., attention (upper alpha), working memory (theta), and suppression of non-essential or potentially interfering neural activity (delta and lower alpha). However, since these effects were not associated with improved performance they may represent compensatory responses required to maintain performance at baseline (pre-treatment) levels in response to fasting. Although response times to the selection of answers were improved in *fasting* children—an effect that could have been influenced by increased arousal associated with fasting [77]—this performance measure involves processes subsequent to those involved in mental calculation, such as long-term memory and visual search [78].

The emphasis on memory-related functions for theta and inhibitory functions for delta and alpha in the interpretation of the results provides a plausible, parsimonious integration of treatment-related EEG and behavioral findings. However, these frequency bands have other functional correlates that could be meaningfully related to processes underlying mathematical thinking. For example, theta activity has been shown to increase during the processing of semantic information [35] and, since mathematical thinking also relies on semantic processes [79], the observed treatment-related increases in theta bands may in part reflect semantic information processing correlates of this activity. Both delta and alpha synchronization have also been related to processes other than inhibition that could be contributing to the increases in these frequency bands across test phases. Increased delta activity has been associated with the activation of working memory processes [30,73], and increased alpha with high internal processing demands [80], with working memory [81], and access to memory traces [48]. Although it is not possible to disentangle the contributions these various correlates may make to the findings, the conclusion remains: children who continue fasting require greater activation of processes associated with these EEG frequencies than those who have eaten breakfast to maintain performance accuracy at the Phase 1 level.

In terms of the contribution of specific meal nutrients to the study results, an influential role for carbohydrates is suggested by the significant group difference in blood glucose at the start of Phase 2 testing. Blood-borne glucose is considered to be the most important energy source for the central nervous system [82–84] and is fundamental to cognitive activity [14,49] including mental arithmetic [12]. Administration of the math paradigm occurred within a post-meal time window when beneficial behavioral effects of increased blood glucose levels are likely to be present since levels had not significantly declined at the time the third sample was taken 2 h after the breakfast meal.

4.4. Other dietary factors

Other dietary factors that may have influenced the results include nutrient–nutrient interactions (e.g., carbohydrate to protein balance) and their effects on synthesis of brain neurotransmitters. Meals rich in carbohydrates increase serotonin synthesis while those rich in protein increase dopamine and norepinephrine synthesis [85]. These neurotransmitter systems are involved in the regulation of functions important for mental arithmetic such as attention and memory processes [86,87], often, however, with oppositional effects. For example, alertness is facilitated by greater levels of dopamine but may be decreased by increased levels of serotonin [88]. Furthermore, small amounts of protein in a meal can attenuate the serotonin-promoting effects of carbohydrates [89]. In the present study facilitating effects on mathematical cognition were observed following a breakfast meal that contained proportionately more carbohydrates than proteins (~4:1 ratio). In a study in adults, Fischer and colleagues [90] reported improved attention and decision times within the first hour after eating a breakfast meal containing a similar balance of carbohydrates-to-proteins. These effects were not accompanied by increases in blood levels of the serotonin precursor tryptophan which was presumably blocked by the protein content of the meal. This would serve to promote the cognitive enhancing effects of food-induced glucose on brain function. It is plausible that a similar effect contributed to the positive neurocognitive effects observed in this study.

5. Caveats and conclusions

The results indicate a coherent set of effects on neural networks involved in mental arithmetic related to the presence or absence of morning nutrition in children, but there are aspects of the study design that may have served to attenuate treatment differences. For example, greater differences between *fed* and *fasting* children might have been observed if the breakfast meal (340 cal) had been more substantial. The USDA School Breakfast Program recommends a minimum 554 cal meal for 8–11 year old children. Furthermore, the influence of meal size may have been compounded by having children eat at mid-morning—

thereby increasing pre-breakfast fasting influences. It could also be that the addition problems presented may not have been challenging enough to reveal greater treatment-related performance differences. In this regard, it is not possible to determine the extent to which the observed error rates that averaged 23% across groups and test phases reflected difficulties with calculation independent of the influence of the demanding time-related features of the task, i.e., brief time periods for viewing problems and answers (200 ms each) and for selecting answers (1.75 s after offset of answer options).

The above considerations point out aspects of the study design that might have been modified to more effectively demonstrate the effects of morning nutrition on a complex cognitive function in children. However, the study design was strengthened by features that, if left uncontrolled, could significantly impact brain function and performance independent of nutritional status. These include: documenting sleep amount and quality the night before morning testing to ensure study measures were not compromised by the effects of sleep deprivation [91]; conducting morning sessions according to a prescribed schedule that controlled for time-of-day influences and intervals between dietary manipulations and assessments; and maintaining dietary and physical activity control the evening and morning before assessments.

In summary, the results suggest that eating breakfast facilitates brain processes involved in mathematical thinking in children. This conclusion is supported by findings showing that during mental calculation neural network activity involved in processing numerical information is functionally enhanced and related to improved performance in children who have eaten breakfast, and that greater mental effort is required for this mathematical thinking in children who skip breakfast. However, the time window used in these analyses encompassed multiple processes involved in mathematical calculations, e.g., the perception and recognition of digits, the retrieval or calculation of the answer, and the storage of this answer in memory in preparation of selecting the correct answer from among those presented. It will be important for future studies to determine the extent to which these different stages of mathematical information processing may be selectively influenced by morning nutrition in children.

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